



**UTM**  
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# Project Proposal (WorkBeacon)

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|                     |                                 |
|---------------------|---------------------------------|
| <b>Programme</b>    | : Bachelor of Computer Science  |
| <b>Subject Code</b> | : SECJ3623                      |
| <b>Subject Name</b> | : PENGATURCARAAN APLIKASI MOBIL |
| <b>Session-Sem</b>  | : 2025/2026-01                  |

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# Project Background and Problem Statement

In high-risk workplace environments such as manufacturing plants, construction sites, or industrial operations, effective communication between management and on-ground staff is critical to ensuring safety and minimizing operational risks. Health and Safety (H&S) teams are required to respond quickly to incidents, disseminate safety alerts, and ensure compliance with safety procedures. However, many organizations still rely on fragmented communication methods such as messaging applications, phone calls, paper-based reporting, or standalone systems that are not designed for real-time operational use.

At Fulkrum Interactive Technology, the need for a complementary mobile solution was identified through discussions with Nur Hazimah, the HR Lead who also oversees Health and Safety matters. As a key stakeholder working closely with the Product Owner, Nur Hazimah highlighted challenges faced by Health and Safety departments when supporting clients operating in high-risk workplace environments.

From a stakeholder perspective, the main challenges include:

- Difficulty in broadcasting urgent safety alerts to all relevant staff in real time.
- Lack of visibility on whether safety alerts have been received and acknowledged by workers.
- Inefficient incident reporting processes that delay response and escalation.
- Absence of a centralized system to track incidents, responses, and resolutions.
- Limited ability for H&S teams to monitor safety compliance during daily operations.

To address these challenges, WorkBeacon is proposed as a complementary mobile application that will be developed alongside a client's main system. The app will primarily support the client's Health and Safety department during daily operations in high-risk environments. By focusing on real-time alerts, acknowledgment tracking, and structured incident management, WorkBeacon aims to enhance workplace safety, improve response times, and provide operational transparency for stakeholders involved in safety management.

# Objectives and Scope

## 2.1 Project Objectives

The objectives of the WorkBeacon application are as follows:

1. To provide a real-time communication platform for workplace alerts and announcements.
2. To enable staff members to report workplace incidents quickly and efficiently.
3. To allow management and HR teams to monitor staff acknowledgment and response status.
4. To improve workplace safety through structured incident tracking and resolution.
5. To provide centralized records for reporting, auditing, and analysis of workplace incidents.

## 2.2 Project Scope

The scope of the WorkBeacon project includes:

- Development of a mobile application for staff and administrators.
- Implementation of two main user roles: Admin (HR/Management) and Staff.
- Real-time alert broadcasting and acknowledgment tracking.
- Incident reporting, management, and status tracking.
- Basic reporting and dashboard views for administrators.

The project does not cover advanced analytics, external emergency service integration, or hardware-based alert systems in its initial phase.

# Target Users and User Needs

## 3.1 Target Users

1. Administrators (HR / Management)
  - Responsible for workplace communication, safety monitoring, and incident management.
2. Staff Members
  - Employees who receive alerts, acknowledge notifications, and report incidents.

## 3.2 User Needs

| User Type | User Needs                                       |
|-----------|--------------------------------------------------|
| Admin     | Broadcast urgent alerts to all or selected staff |
| Admin     | Track who has acknowledged alerts                |
| Admin     | View, manage, and resolve reported incidents     |
| Admin     | Access reports and incident history              |
| Admin     | Track staff live locations                       |
| Staff     | Receive real-time alerts and announcements       |
| Staff     | Acknowledge alerts quickly                       |
| Staff     | Report incidents with clear descriptions         |
| Staff     | Track the status of submitted incidents          |

## Proposed Features

The proposed features of WorkBeacon are specifically designed to address the operational challenges faced by Nur Hazimah and the Health and Safety teams of client organizations.

### 4.1 Features for Health & Safety Admin (HR / Management)

| Feature                       | Stakeholder Problem Addressed                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------|
| Safety Alert Broadcasting     | Enables HR/H&S leads to quickly disseminate urgent safety notices to staff working in hazardous areas |
| Alert Acknowledgment Tracking | Allows Nur Hazimah and H&S teams to verify which staff have received and acknowledged safety alerts   |
| Incident Management Dashboard | Provides centralized visibility of all reported incidents, their status, and response progress        |
| Incident Status Updates       | Ensures accountability by tracking incidents from reporting to resolution                             |
| Activity Logs & Records       | Supports auditing, compliance checks, and safety performance reviews                                  |
| User & Role Management        | Ensures only authorized personnel can issue safety alerts or manage incidents                         |

### 4.2 Staff Features (Operational Workers)

| Feature                  | Stakeholder Problem Addressed                                                         |
|--------------------------|---------------------------------------------------------------------------------------|
| Real-Time Notifications  | Ensures staff receive immediate alerts related to safety risks or operational hazards |
| Alert Acknowledgment     | Confirms that safety messages are read and understood                                 |
| Incident Reporting       | Enables quick reporting of hazards, near-misses, or accidents from the worksite       |
| Incident Status Tracking | Keeps staff informed on the progress of reported safety issues                        |

### 4.3 System Features

- Secure authentication with role-based access control
- Push notification support for urgent safety alerts
- Centralized data storage for incidents and alerts
- Designed as a supporting mobile system
- Optimized for use in high-risk operational environments

### Project Timeline and Milestones

| Phase   | Activities                           | Estimated Duration |
|---------|--------------------------------------|--------------------|
| Phase 1 | Requirement Analysis & Planning      | Week 1 – 2         |
| Phase 2 | System Design (UI/UX & Architecture) | Week 3 – 4         |
| Phase 3 | Application Development              | Week 5 – 8         |
| Phase 4 | Testing & Bug Fixing                 | Week 9 – 10        |
| Phase 5 | Deployment & Evaluation              | Week 11 – 12       |

### Conclusion

WorkBeacon aims to enhance workplace communication and safety by providing a centralized, real-time mobile platform for alerts and incident management. By improving response tracking, transparency, and documentation, the application supports organizations in creating a safer and more efficient working environment. This project also demonstrates practical application of mobile application development concepts, including user-centered design, role-based access control, and real-time communication systems.